Lecture 13: **Nuclear reactions and applications**

MODERN PHYSICS

DR. SZILVÁSI RÉKA

ASSISTANT PROFESSOR

BME NTI

Outline

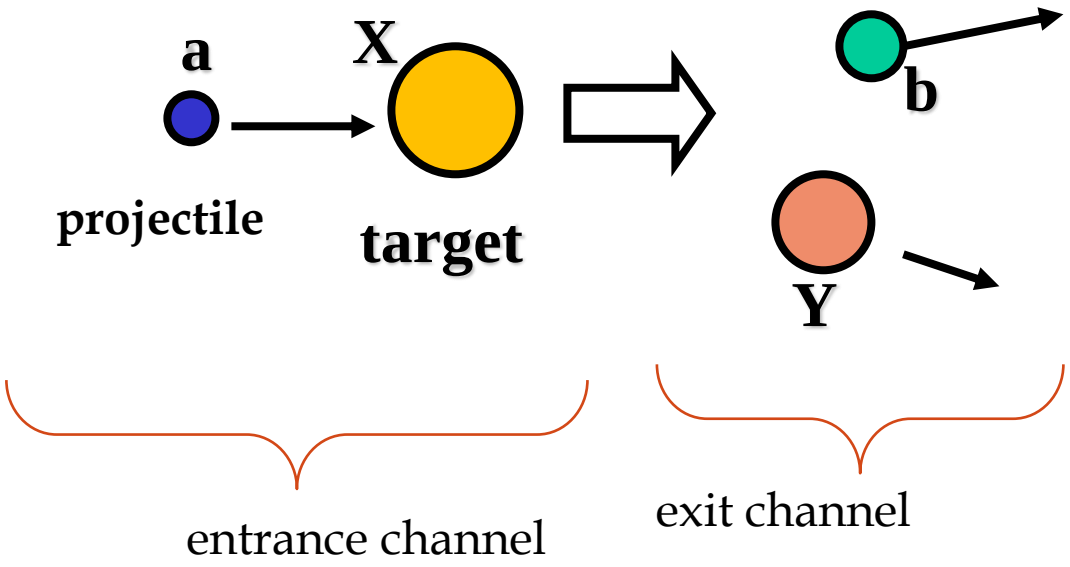
- Nuclear reaction **kinematics, accelerators**
- The **cross section** and statistical properties
- **Neutron-activated** reactions: nuclear fission and nuclear fusion
- Fundamentals of **fission** reactors
- Fundamentals of **fusion** reactors
- Applications of radiation in different fields

Nuclear reactions

KINEMATICS AND STATISTICAL QUANTITIES

Conventions

Remember: First nuclear reaction produced by Rutherford: $\alpha + {}^{14}_{7}\mathrm{N} \rightarrow p + {}^{17}_{8}\mathrm{O}$



$$a + X \longrightarrow Y + b$$

$$X(a, b)Y$$

- The (a,b) reaction of nucleus X!

- A given initial state may lead to different final states!

Conserved quantities

Space-time symmetries:

- **Energy** (time translation symmetry)
- **Momentum** (space translation symmetry)
- **Angular momentum** (rotational symmetry)

Internal symmetries (charges)

- **Electric charge**
- **Barionic charge** (**nucleon number**)
- **Leptonic charge** (light fermion number)

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Internal symmetries (charges)

- **Electric charge**
- **Barionic charge** (nucleon number)
- **Leptonic charge** (light fermion number)
- **Parity** (spatial coordinate flips)

Energy conservation:

The process: $a + b \rightarrow c + d$ ⇨ $E_{initial} = E_{final}$

$$M_a c^2 + T_a + M_b c^2 + T_b = M_c c^2 + T_c + M_d c^2 + T_d$$

rest energy

Kinetic energy

Reaction energy

$$M_a c^2 + T_a + M_b c^2 + T_b = M_c c^2 + T_c + M_d c^2 + T_d$$

Independent of the reference frame!

Depends on the reference frame!

Characterizing the reaction: reorder the above to a quantity independent of the observer

Reaction energy:

$$Q = M_a c^2 + M_b c^2 - M_c c^2 - M_d c^2$$

$$M_a c^2 = Z_a m_p c^2 + (A - Z)_a m_n c^2 - B_a \quad !$$

- for endothermic reactions: $\boldsymbol{Q < 0}$ $\quad Q = T_c + T_d - (T_a + T_b) < 0$

- remember that the **kinetic energy is positive definit:**

$\geq 0 \qquad \geq 0$

$-Q < (T_a + T_b)$

- $Q + (T_a + T_b) = T_c + T_d \geq 0$

For an endothermic reaction to proceed, the incident particle must have a minimum kinetic energy called the **threshold energy**!

Since it is determined by the kinetic energies, **the threshold energy of the reaction depends on the reference frame**!

- The expression for the threshold energy is true in the CM frame only…

$$-Q < \left(T_{a,CM} + T_{b,CM}\right) = T_{CM}tresh$$

- **In laboratory frame**, one must transform the system
  (homework)

$$T^{l}_{tresh} = Ta \geq -Q\left(1 + \frac{M_a}{M_b}\right)$$

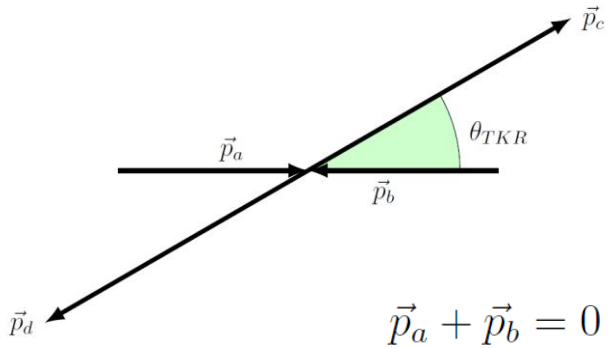
  - projectile: **a**
  - target: **b at rest**
  - T_a **is higher than** $T_{a,CM}$

  - for: Ma>>Mb ⟶ significant difference
    - Example:
      $^{1}H(\alpha, \alpha')^{1}H$

  - for: Ma<<Mb (heavy target) ⟶ $\boldsymbol{T^{l}_{tresh} \approx T_{CM}tresh}$

Kinematics

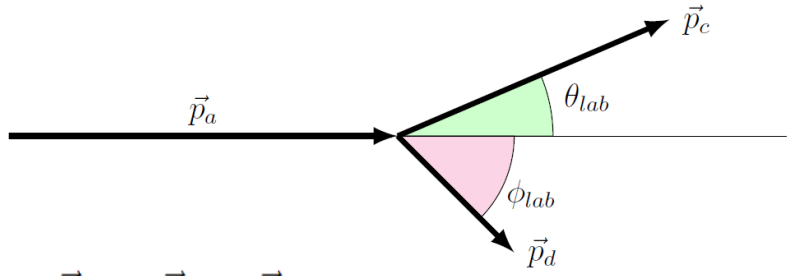
CM frame



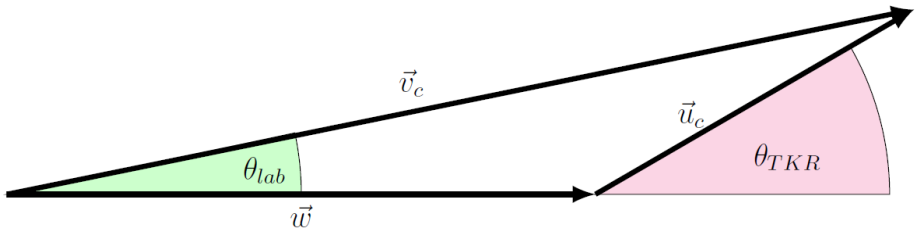
$$\vec{p}_a + \vec{p}_b = 0$$
$$\vec{p}_c + \vec{p}_d = 0$$

$$\vec{u}_a = \vec{v}_a - \vec{w}$$
$$\vec{u}_b = \vec{v}_b - \vec{w}$$
$$\vec{u}_c = \vec{v}_c - \vec{w}$$
$$\vec{u}_d = \vec{v}_d - \vec{w}$$

Laboratory frame



$$\vec{p}_a = \vec{p}_c + \vec{p}_d$$



$$u_c \cdot \sin\theta_{TKR} = v_c \cdot \sin\theta_{lab}$$

$$\sin\theta_{lab} \leq \frac{u_c}{v_c}$$

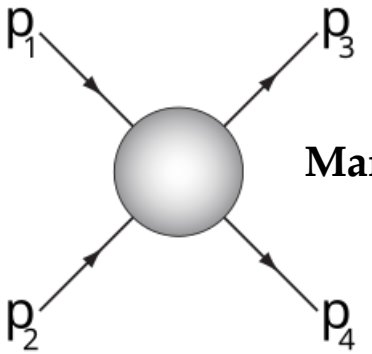
- If $u_c < v_c$ there is a maximum laboratory scattering angle!

Special situation: high-energy colliding beams

Consider 2-particle collisions with relativistic energies

in a collision the **centre-of-mass energy** defined by the *4-momenta* of the incoming particles:

$$E_{CM} = \sqrt{s}.$$



Mandelstam variables

$$s = (p_1 + p_2)^2 = \mathrm{M}^2$$
$$t = (p_1 - p_3)^2$$
$$u = (p_1 - p_4)^2$$

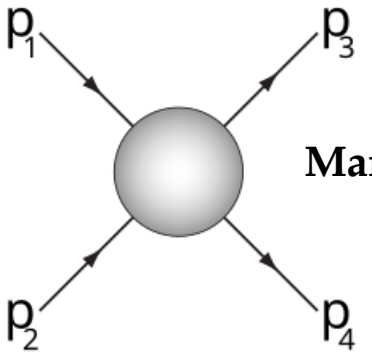
$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2.$$

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Remember:

$p_i^2 = m_i^2$ invariant mass

$m^2 = E^2 - \|\mathbf{p}\|^2$

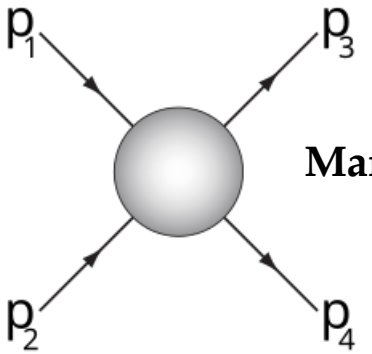
$p_1=(E_1,\mathbf{p_1})$ $p_2=(E_2,\mathbf{p_2})$

Special situation: high-energy colliding beams

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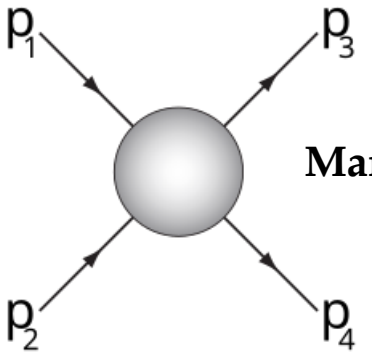
$$M^2 = (E_1 + E_2)^2 - \|\mathbf{p}_1 + \mathbf{p}_2\|^2$$
$$= m_1^2 + m_2^2 + 2\,(E_1 E_2 - \mathbf{p}_1 \cdot \mathbf{p}_2)$$

Special situation: high-energy colliding beams

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Remember:

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$m^2 = E^2 - \|\mathbf{p}\|^2$

In CM frame: $0 = \sum_i \vec{p}_i$

$$M^2 = (E_1 + E_2)^2 - \|\mathbf{p}_1 + \mathbf{p}_2\|^2 = m_1^2 + m_2^2 + 2\,(E_1 E_2 - \mathbf{p}_1 \cdot \mathbf{p}_2)$$

$$p_1 = (E_p, \vec{p})$$
$$p_2 = (E_p, -\vec{p}).$$

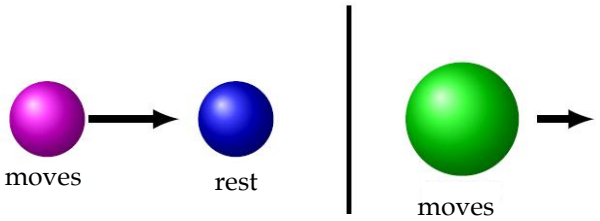
$$s = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = (E_p^2 - |\vec{p}|^2) + (E_p^2 - |\vec{p}|^2) + 2(E_p^2 + |\vec{p}|^2) = 4E_p^2$$

Special situation: high-energy colliding beams

Inelastic scattering for the creation of particles: *E_{CM} can be used for that?* $p + p \rightarrow p + p + p + \bar{p}$

$$a + b \rightarrow c$$

1. **Target at rest case:**
   (LAB frame)



$$E_{\mathrm{cm}} = \sqrt{(m_1c^2 + m_2c^2)^2 + 2m_2c^2K}$$

Kinetic energy of the projectile

2. **Colliding beams:**

$$p_1 = (E_p, \vec{p}) \longrightarrow \quad \longleftarrow p_2 = (E_p, -\vec{p}).$$

$$0 = \sum_i \vec{p}_i$$

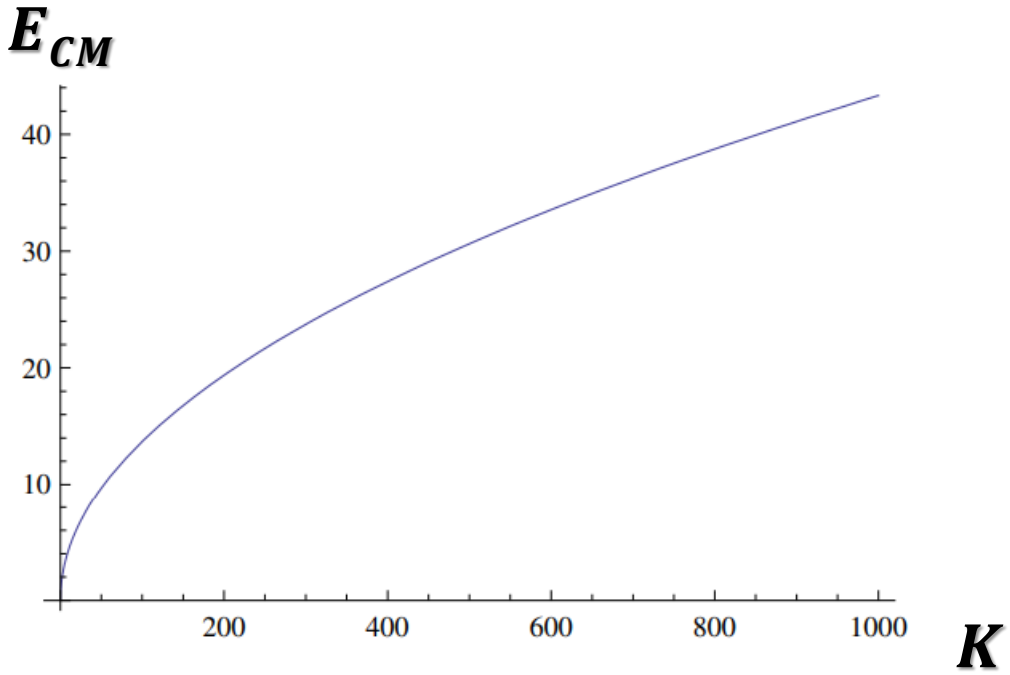
$$E_{CM} = 2 \times Ep$$

The amount of energy available for the creation of particles is higher in this case!

$\overrightarrow{22\,\mathrm{GeV}} + \overleftarrow{22\,\mathrm{GeV}}$ has the same E_{CM} as $\overrightarrow{1\,\mathrm{TeV}} + m_{\mathrm{target}}$;

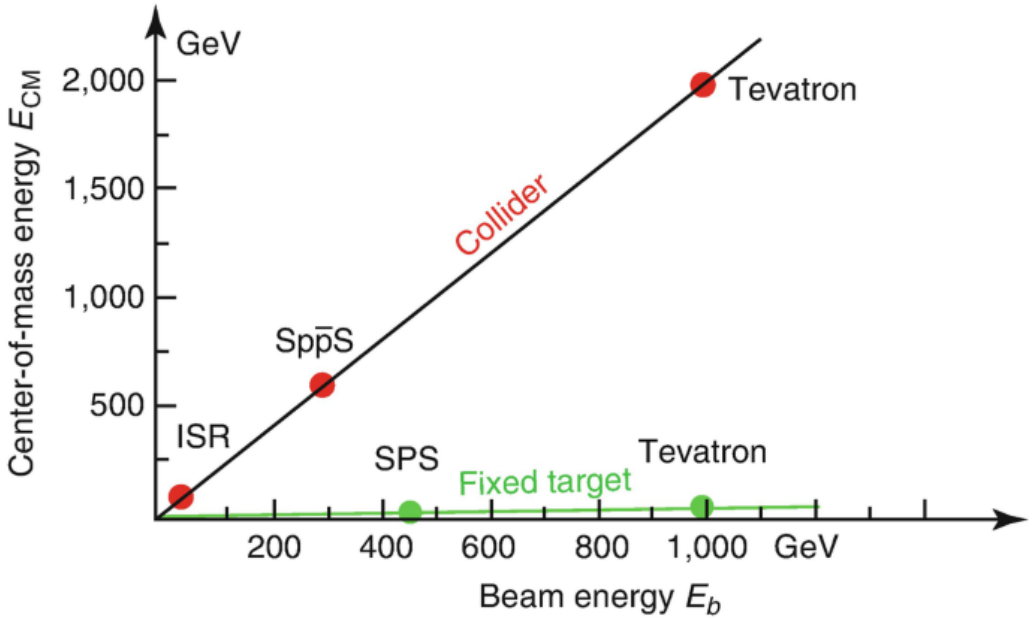
$\overrightarrow{1\,\mathrm{TeV}} + \overleftarrow{1\,\mathrm{TeV}}$ has the same E_{CM} as $\overrightarrow{10^3\,\mathrm{TeV}} + m_{\mathrm{target}}$.

$$E_{\mathrm{cm}} = \sqrt{(m_1c^2 + m_2c^2)^2 + 2m_2c^2K}$$



Fixed target case

$$E_{CM} = 2 \times Ep$$



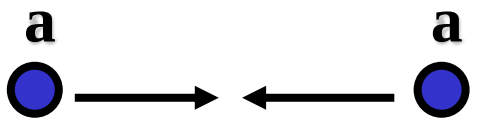
Colliding beams case

Special situation: high-energy colliding beams

- **equal-mass** and **opposite-momentum** particles
- colliding **head-on**

- **cyclotron**

CM frame equals the LAB frame!



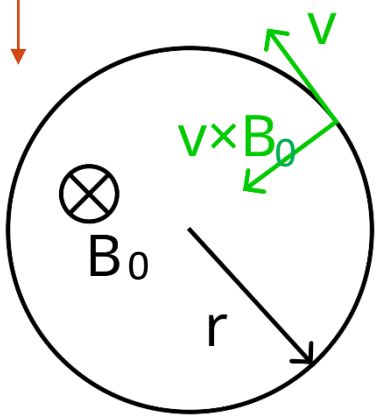
$$E_{CM} = 2 \times Ep$$

$$0 = \sum_i \vec{p}_i$$

Special situation: high-energy colliding beams - acceleration

- **equal-mass** and **opposite-momentum** particles
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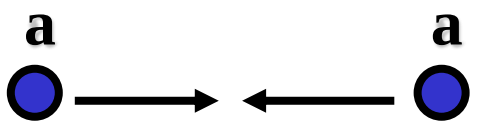
$$m\mathbf{a} = q\mathbf{v} \times \mathbf{B}_0$$

$$m\frac{v^2}{r} = qvB_0$$

$$r = \frac{mv}{qB_0}$$

cyclotron radius

CM frame equals the LAB frame!



$$\boldsymbol{E_{CM} = 2 \times Ep}$$

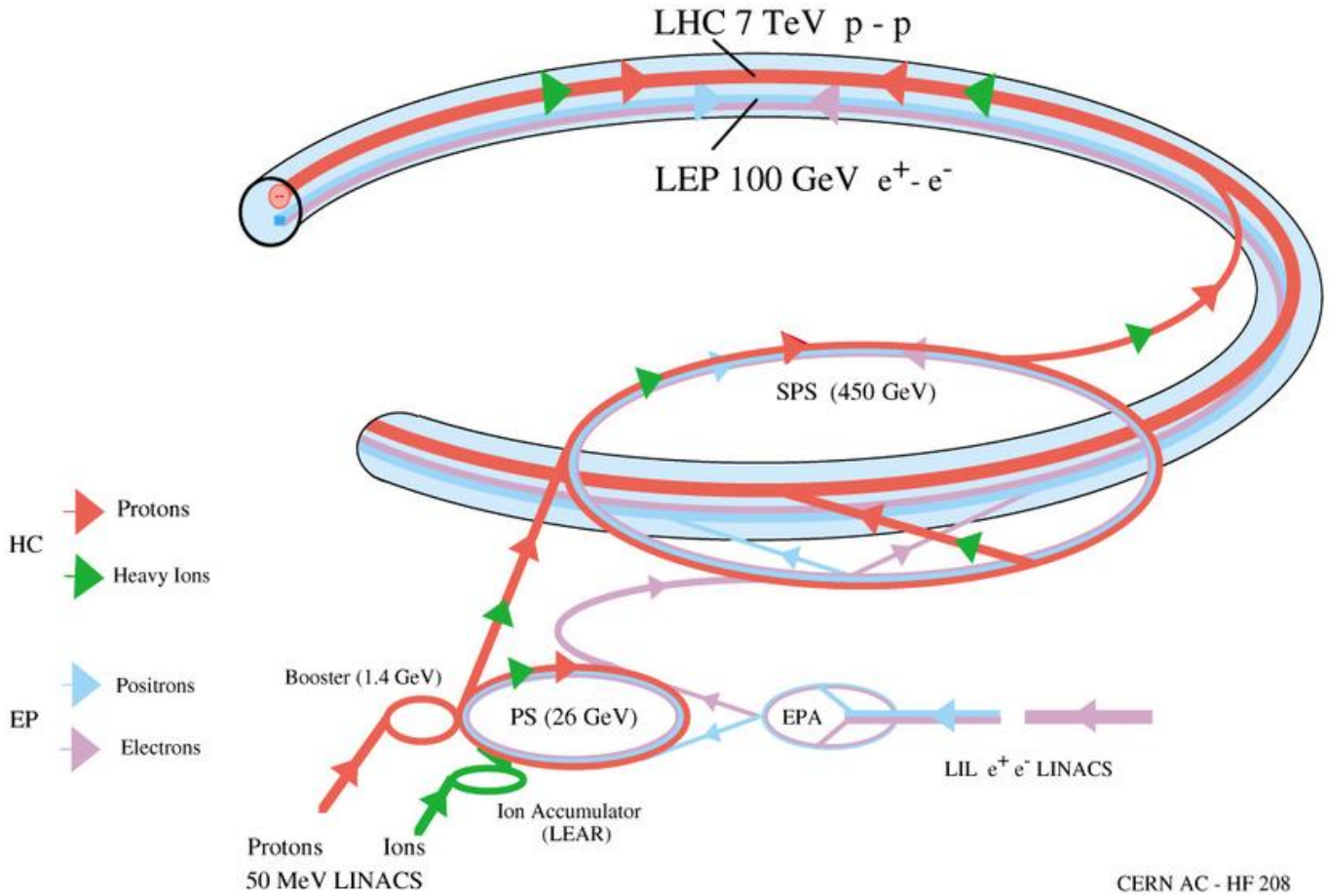
$$0 = \sum_i \vec{p}_i$$

Circular path by **magnetic field** + an **accelerating electric field** (every time particles run a full circle they pass through the accelerating electrodes) → $\mathbf{\Delta E_{kin} = qU}$

Special situation: high-energy colliding beams - acceleration

- LEP → LHC
- *100 GeV leptons* *7 TeV protons*

- how much stronger magnetic field is needed to keep the particles on a circular path of the same radius as in the LEP?



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$$E = \sqrt{m^2c^4 + p^2c^2} \approx pc.$$

$$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{E}{eBc},$$

$$\frac{B_{\mathrm{LHC}}}{B_{\mathrm{LEP}}} = \frac{E_{LHC}}{E_{LEP}} = \frac{7000\,\mathrm{GeV}}{101\,\mathrm{GeV}} = 69.3$$

$A = \pi r^2$

